TRPO & PPO

Samsung AI Expert

Jul 10, 2025

Giho Kim

Learning and Decision Systems

TRPO - Review

Learning and Decision Systems

TRPO-Review : 1) Surrogate Objective

$$\max_{\pi} \; J(\pi) - J(\pi_{old}) \;\; = \sum_t \mathbb{E}_{s_t \sim p^{\pi}} \left[\mathbb{E}_{a_t \sim \pi(\cdot|s_t)} \left[\gamma^t A^{\pi_{old}}(s_t, a_t) \right] \right]$$

What is $A^{\pi}(s_t, a_t)$?
$Q^{\pi}(s_t, \mathbf{a_t}) - V^{\pi}(s_t)$

LDS
Learning and Decision Systems

TRPO-Review : 1) Surrogate Objective

$$\max_{\pi} \; J(\pi) - J(\pi_{old}) \; = \sum_t \mathbb{E}_{s_t \sim p^{\pi}} \left[\mathbb{E}_{a_t \sim \pi(\cdot|s_t)} \left[\gamma^t A^{\pi_{old}}(s_t, a_t) \right] \right]$$

$$= \sum_t \mathbb{E}_{s_t \sim p^{\pi}} \left[\mathbb{E}_{a_t \sim \pi_{old}(\cdot|s_t)} \left[\frac{\pi(a_t|s_t)}{\pi_{old}(a_t|s_t)} \gamma^t A^{\pi_{old}}(s_t, a_t) \right] \right] \quad \text{(importance sampling)}$$

What is importance sampling?

$$\mathbb{E}_{p(x)}[f(x)] = \int_x p(x) f(x) dx = \int_x q(x) \frac{p(x)}{q(x)} f(x) dx = \mathbb{E}_{\mathbf{q}(\mathbf{x})}[\frac{p(x)}{q(x)} f(x)]$$

Learning and Decision Systems

TRPO-Review : 1) Surrogate Objective

$$\max_{\pi} \; J(\pi) - J(\pi_{old}) \;\; = \sum_t \mathbb{E}_{s_t \sim p^{\pi}} \left[\mathbb{E}_{a_t \sim \pi(\cdot|s_t)} \left[\gamma^t A^{\pi_{old}}(s_t, a_t) \right] \right]$$

$$= \sum_t \mathbb{E}_{s_t \sim p^{\pi}} \left[\mathbb{E}_{a_t \sim \pi_{old}(\cdot|s_t)} \left[\frac{\pi(a_t|s_t)}{\pi_{old}(a_t|s_t)} \gamma^t A^{\pi_{old}}(s_t, a_t) \right] \right] \quad \text{(importance sampling)}$$

$$\geq L_{\pi_{old}}(\pi) \; - C \max_s D_{KL}(\pi_{old}(\cdot|s) \parallel \pi(\cdot|s)) \quad \text{(surrogate objective)}$$

$$\boxed{L_{\pi_{old}}(\pi) := \sum_t \mathbb{E}_{s_t \sim p^{\pi_{old}}} \left[\mathbb{E}_{a_t \sim \pi_{old}(\cdot|s_t)} \left[\frac{\pi(a_t|s_t)}{\pi_{old}(a_t|s_t)} \gamma^t A^{\pi_{old}}(s_t, a_t) \right] \right]}$$

Learning and Decision Systems

TRPO-Review : 1) Surrogate Objective

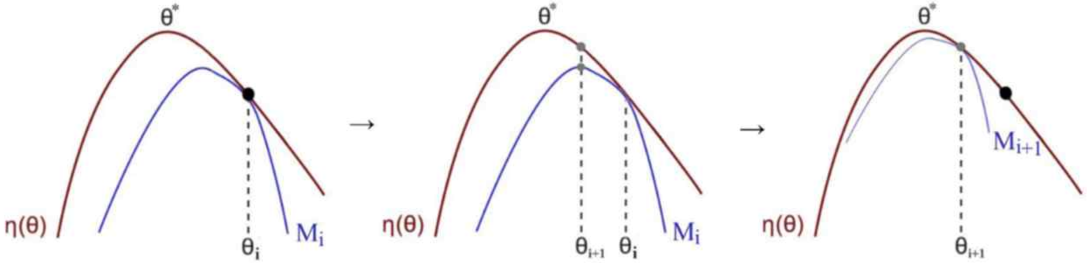
$$\max_{\pi} \; J(\pi) - J(\pi_{old}) = \sum_t \mathbb{E}_{s_t \sim p^{\pi}} \left[\mathbb{E}_{a_t \sim \pi(\cdot|s_t)} \left[\gamma^t A^{\pi_{old}}(s_t, a_t) \right] \right]$$

$$= \sum_t \mathbb{E}_{s_t \sim p^{\pi}} \left[\mathbb{E}_{a_t \sim \pi_{old}(\cdot|s_t)} \left[\frac{\pi(a_t|s_t)}{\pi_{old}(a_t|s_t)} \gamma^t A^{\pi_{old}}(s_t, a_t) \right] \right] \quad \text{(importance sampling)}$$

$$\geq L_{\pi_{old}}(\pi) - C \max_s D_{KL}(\pi_{old}(\cdot|s) \,\|\, \pi(\cdot|s)) \quad \text{(surrogate objective)}$$

Requirements?

- $M(\theta|\theta_i) \leq f(\theta) \quad \forall \theta$
- $M(\theta_i) = f(\theta_i)$



TRPO-Review : 1) Surrogate Objective

$$\max_\pi J(\pi) - J(\pi_{old}) = \sum_t \mathbb{E}_{s_t \sim p^\pi}\left[\mathbb{E}_{a_t \sim \pi(\cdot|s_t)}\left[\gamma^t A^{\pi_{old}}(s_t, a_t)\right]\right]$$

$$= \sum_t \mathbb{E}_{s_t \sim p^\pi}\left[\mathbb{E}_{a_t \sim \pi_{old}(\cdot|s_t)}\left[\frac{\pi(a_t|s_t)}{\pi_{old}(a_t|s_t)}\gamma^t A^{\pi_{old}}(s_t, a_t)\right]\right] \quad \text{(importance sampling)}$$

$$\geq L_{\pi_{old}}(\pi) - C \max_s D_{KL}(\pi_{old}(\cdot|s) \,\|\, \pi(\cdot|s)) \quad \text{(surrogate objective)}$$

[1]where $C = 4\epsilon\gamma/(1-\gamma)$, $\epsilon = \max_{s,a}|A_\pi(s,a)|$

[1]Schulman, John, et al. "Trust region policy optimization." *International conference on machine learning*. PMLR, 2015.

LDS
Learning and Decision Systems

TRPO-Review : 2) Approximate trust region

$$\boxed{\max_{\theta} L_{\theta_{old}}(\theta) - C \max_{s} D_{KL}(\theta_{old} \parallel \theta)}$$

too small stepsize

TRPO-Review : 2) Approximate trust region

$$\max_{\theta} L_{\theta_{old}}(\theta) - C \max_{s} D_{KL}(\theta_{old} \parallel \theta)$$

too small stepsize

$$\begin{aligned} &\underset{\theta}{\text{maximize}}\, L_{\theta_{\text{old}}}(\theta) \\ &\text{subject to } D_{\text{KL}}^{\max}(\theta_{\text{old}}, \theta) \leq \delta. \end{aligned}$$

trust region constraint

Learning and Decision Systems

TRPO-Review : 2) Approximate trust region

$$\max_{\theta} L_{\theta_{old}}(\theta) - C \max_{s} D_{KL}(\theta_{old} \,\|\, \theta)$$

too small stepsize

$$\underset{\theta}{\text{maximize}}\; L_{\theta_{\text{old}}}(\theta)$$
$$\text{subject to } D_{\text{KL}}^{\max}(\theta_{\text{old}}, \theta) \le \delta.$$

trust region constraint

bounded at every point in the state space

TRPO-Review : 2) Approximate trust region

$$\max_{\theta} L_{\theta_{old}}(\theta) - C \max_{s} D_{KL}(\theta_{old} \parallel \theta)$$

too small stepsize

$$\underset{\theta}{\text{maximize}}\ L_{\theta_{\text{old}}}(\theta)$$
$$\text{subject to } D_{\text{KL}}^{\max}(\theta_{\text{old}}, \theta) \le \delta.$$

trust region constraint

$$\begin{aligned} \max_{\theta} \quad & L_{\theta_{old}}(\theta) \\ \text{s.t.} \quad & \underbrace{\mathbb{E}^{\pi_{\theta_{old}}}[D_{KL}(\pi_{\theta_{old}} \parallel \pi_{\theta})] \le \delta}_{\text{approximate trust region}} \end{aligned}$$

approximation (max. $\rightarrow$ avg.)

Learning and Decision Systems

TRPO-Implementation

TRPO-Implementation

What info do we need when we implement **on-policy algorithms**?

1. (s_j, a_j, s'_j, r_j)
2. generalized advantage estimation(GAE)
3. MC estimates of value $V^{\pi_\phi}(s)$
4. probability $\pi_{\phi_{\text{old}}}(a_j|s_j)$
5. policy & value update

Learning and Decision Systems

TRPO-Implementation

policy gradient (with importance sampling)

$$\nabla_\phi J(\phi) = \mathbb{E}_{s\sim\rho_{\phi\text{old}},\ a\sim\pi_{\phi_\text{old}}} \left(A^{\pi_\phi}(s,a) \frac{\nabla_\phi \pi_\phi(a|s)}{\pi_{\phi_\text{old}}(a|s)} \right)$$

To estimate A^{π_ϕ}, we use a separate value function approximator $V(s;\theta)$, and apply **generalized advantage estimation(GAE)**!

Furthermore, we use a stochastic policy $\pi_\phi(a|s)$ (Gaussian in our case).

TRPO-Implementation

GAE?

Given a trajectory $(s_0, a_0, r_0, s_1, a_1, r_1, \cdots s_T)$ generated by executing the current policy, we first compute

$$\delta_t = r_t + \gamma V(s_{t+1};\theta) - V(s_t;\theta)$$

Then, GAE is computed as follows:

$$\hat{A}(s_t, a_t) = \sum_{l=0}^{\infty} (\lambda\gamma)^l \delta_{t+l}$$

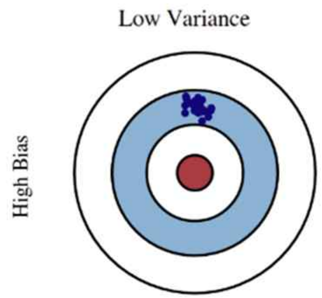
Learning and Decision Systems

TRPO-Implementation

Generalized Advantage Estimation

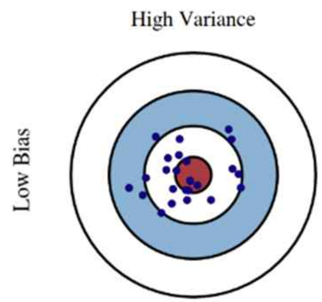
- $\lambda = 0$

$$\hat{A}(s_t, a_t) = r_t + \gamma V(s_{t+1}; \theta) - V(s_t; \theta)$$



- $\lambda = 1$

$$\hat{A}(s_t, a_t) = r_t + r_{t+1} + r_{t+2} + \ldots$$



Learning and Decision Systems

TRPO-Implementation

training $V(s;\theta)$?

given a trajectory $(s_0, a_0, r_0, s_1, a_1, r_1, \cdots s_T)$ generated from π_ϕ, we can compute Monte-Carlo esimtates of $V(s_0), V(s_1), \cdots V(s_T)$ as follows:

$$V(s_t) = \sum_{\tau=t}^{T-1} \gamma^{\tau-t} r_\tau + \gamma^T V(s_T)$$

This is the **target** for value function update!

Learning and Decision Systems

TRPO-Implementation


```
self._obs_mem = np.zeros(shape=(lim, dimS))
self._act_mem = np.zeros(shape=(lim, dimA))
self._rew_mem = np.zeros(shape=(lim,))
self._val_mem = np.zeros(shape=(lim,))
self._log_prob_mem = np.zeros(shape=(lim,))
```

collected during agent-env interaction

```
# memory of cumulative rewards which are MC-estimates of the current value function
self._target_v_mem = np.zeros(shape=(lim,))
# memory of GAE($\lambda$)-estimate of the current advantage function
self._adv_mem = np.zeros(shape=(lim,))
```

computed when sampling a single episode is done

Learning and Decision Systems

TRPO-Implementation

$$\max_{\theta} \quad L_{\theta_{old}}(\theta)$$
$$\text{s.t.} \quad \underbrace{\mathbb{E}^{\pi_{\theta_{old}}}[D_{KL}(\pi_{\theta_{old}} \parallel \pi_\theta)] \le \delta}_{\text{approximate trust region}}$$

$\xrightarrow{\text{Taylor approximation}}$

$$\max_{\theta} \quad g^\top(\theta - \theta_{old})$$
$$\text{s.t.} \quad \frac{1}{2}(\theta - \theta_{old})^\top H(\theta - \theta_{old}) \le \delta$$

obj. $\rightarrow$ first-order

const. $\rightarrow$ second-order

$$\text{where } H := \nabla_\theta^2[D_{KL}(\pi_{\theta_{old}} || \pi_\theta)],$$
$$g := \nabla_\theta L_{\theta_{old}}(\theta)|_{\theta=\theta_{old}}$$

Learning and Decision Systems

TRPO-Implementation

$$\max_{\theta} \quad L_{\theta_{old}}(\theta)$$
$$\text{s.t.} \quad \underbrace{\mathbb{E}^{\pi_{\theta_{old}}}[D_{KL}(\pi_{\theta_{old}} \parallel \pi_\theta)] \leq \delta}_{\text{approximate trust region}}$$

$\xrightarrow{\text{Taylor approximation}}$

$$\max_{\theta} \quad g^\top(\theta - \theta_{old})$$
$$\text{s.t.} \quad \frac{1}{2}(\theta - \theta_{old})^\top H(\theta - \theta_{old}) \leq \delta$$

Solution.

$$\theta_{new} = \theta_{old} + \sqrt{\frac{2\delta}{g^\top H^{-1} g}} H^{-1} g$$

Learning and Decision Systems

TRPO-Policy & Value Update

Unfortunately, the parameter updates in TRPO...

1. Fisher-vector product
2. conjugate gradient descent $\rightarrow H^{-1}g$
3. backtracking line search $\rightarrow$ still trust region?

Solution.

$$\theta_{new} = \theta_{old} + \sqrt{\frac{2\delta}{g^\top H^{-1} g}} H^{-1} g$$

TRPO-Implementation

1. Fisher-vector product

```
def fisher_vector_product(v, actor, obs_batch, cg_damping=1e-2):
    v.detach()
    kl = torch.mean(kl_div(actor=actor, old_actor=actor, obs_batch=obs_batch))
    kl_grads = torch.autograd.grad(kl, actor.parameters(), create_graph=True)
    kl_grad = torch.cat([grad.view(-1) for grad in kl_grads])
    kl_grad_p = torch.sum(kl_grad * v)
    Iv = torch.autograd.grad(kl_grad_p, actor.parameters()) # product of Fisher information I and v
    Iv = flatten(Iv)
    return Iv + v * cg_damping
```

Basically, Fish information matrix is **Hessian** of a function.
$\Longrightarrow$ costly (requires n^2 numbers of 2^{nd}-order differentiation)

However, computation of Ax can easily be done using automatic differentiation: use

$$(\nabla_\theta^2 f)v = \nabla_\theta(\underbrace{\nabla_\theta f \cdot v}_{\text{scalar}}).$$

Learning and Decision Systems

TRPO-Implementation

2. conjugate gradient method

```python
def cg(f_Ax, b, actor, obs_batch, cg_iters=10, residual_tol=1e-10):
    p = b.clone()
    r = b.clone()
    x = torch.zeros_like(b)
    rdotr = r.dot(r)
    for i in range(cg_iters):
        z = f_Ax(p, actor, obs_batch)
        v = rdotr / p.dot(z)
        x += v*p
        r -= v*z
        newrdotr = r.dot(r)
        mu = newrdotr/rdotr
        p = r + mu*p
        rdotr = newrdotr
        if rdotr < residual_tol:
            break
    return x
```

What is conjugate gradient for?
$\Longrightarrow$ To efficiently solve $As = g$

- Solve $\frac{1}{2}s^T H s = g^T s$, rather than $Hs = g$
- Search along the conjugate directions

Learning and Decision Systems

TRPO-Implementation

3. backtracking linesearch

```
def backtracking_line_search(...):
    backtrac_coef = 1.0
    alpha = 0.5
    beta = 0.5
    flag = False
    expected_improve = (actor_loss_grad * maximal_step).sum(0, keepdim=True)
    for i in range(10):
        new_params = params + backtrac_coef * maximal_step
        update_model(actor, new_params)
        new_actor_loss = surrogate_loss(actor, adv, states, old_policy.detach(), actions)
        loss_improve = new_actor_loss - actor_loss
        expected_improve *= backtrac_coef
        improve_condition = loss_improve / expected_improve
        kl = kl_div(actor=actor, old_actor=old_actor, obs_batch=states)
        kl = kl.mean()
        if kl < max_kl and improve_condition > alpha:
            flag = True
            break
        backtrac_coef *= beta
    if not flag:
        params = flat_params(old_actor)
        update_model(actor, params)
```

$\theta \longleftarrow \theta + \beta s$ where s : search direction found through CG

check if new policy stays within **trust region**

decrease β and repeat the verification

Learning and Decision Systems

TRPO-Implementation

3. backtracking linesearch

```python
def backtracking_line_search(...):
    backtrac_coef = 1.0
    alpha = 0.5
    beta = 0.5
    flag = False
    expected_improve = (actor_loss_grad * maximal_step).sum(0, keepdim=True)
    for i in range(10):
        new_params = params + backtrac_coef * maximal_step
        update_model(actor, new_params)
        new_actor_loss = surrogate_loss(actor, adv, states, old_policy.detach(), actions)
        loss_improve = new_actor_loss - actor_loss
        expected_improve *= backtrac_coef
        improve_condition = loss_improve / expected_improve
        kl = kl_div(actor=actor, old_actor=old_actor, obs_batch=states)
        kl = kl.mean()
        if kl < max_kl and improve_condition > alpha:
            flag = True
            break
        backtrac_coef *= beta
    if not flag:
        params = flat_params(old_actor)
        update_model(actor, params)
```

$\theta \longleftarrow \theta + \beta s$ where s : search direction found through CG

check if new policy stays within **trust region**

decrease β and repeat the verification

Learning and Decision Systems

TRPO-Implementation

3. backtracking linesearch

```python
def backtracking_line_search(...):
    backtrac_coef = 1.0
    alpha = 0.5
    beta = 0.5
    flag = False
    expected_improve = (actor_loss_grad * maximal_step).sum(0, keepdim=True)
    for i in range(10):
        new_params = params + backtrac_coef * maximal_step
        update_model(actor, new_params)
        new_actor_loss = surrogate_loss(actor, adv, states, old_policy.detach(), actions)
        loss_improve = new_actor_loss - actor_loss
        expected_improve *= backtrac_coef
        improve_condition = loss_improve / expected_improve
        kl = kl_div(actor=actor, old_actor=old_actor, obs_batch=states)
        kl = kl.mean()
        if kl < max_kl and improve_condition > alpha:
            flag = True
            break
        backtrac_coef *= beta
    if not flag:
        params = flat_params(old_actor)
        update_model(actor, params)
```

$\theta \longleftarrow \theta + \beta s$ where s : search direction found through CG

check if new policy stays within **trust region**

decrease β and repeat the verification

Learning and Decision Systems

TRPO-Implementation

getting together..

```
g = torch.cat([grad.view(-1) for grad in loss_grad]).data
s = cg(fisher_vector_product, g, agent.pi, states)

sAs = torch.sum(fisher_vector_product(s, agent.pi, states) * s, dim=0, keepdim=True)
step_size = torch.sqrt(2 * delta / sAs)[0]
step = step_size * s
```

$$\boxed{\theta_{new} = \theta_{old} + \sqrt{\frac{2\delta}{g^\top H^{-1} g}} H^{-1} g}$$

Learning and Decision Systems

TRPO-Implementation

getting together..

```
g = torch.cat([grad.view(-1) for grad in loss_grad]).data
s = cg(fisher_vector_product, g, agent.pi, states)

sAs = torch.sum(fisher_vector_product(s, agent.pi, states) * s, dim=0, keepdim=True)
step_size = torch.sqrt(2 * delta / sAs)[0]
step = step_size * s
```

Why?

- $s \approx H^{-1}g$
- The denominator aims to measure $s^\top Hs$ with s being used in the algorithm rather than the theoritical $H^{-1}g$.

TRPO- Complete Implementation

Learning and Decision Systems

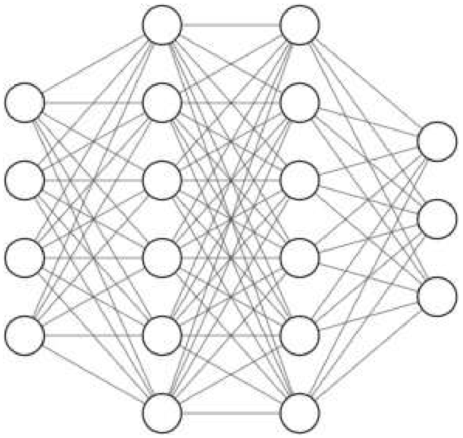
TRPO-Network Arichitecture

Key components of TRPO agent:

- policy network
  - can handle both discrete/continuous actions
  - network ouput : **probability distribution** over action space (cf. DDPG)

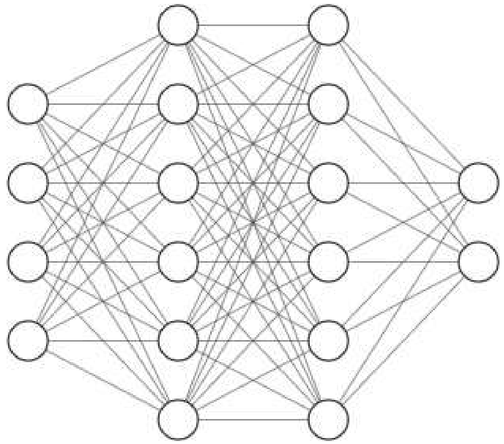
ex) discrete action space

$s \Longrightarrow$



ex) continuous action space

$s \Longrightarrow$



$a \sim \mathcal{N}(\mu(s), \sigma(s))$

Learning and Decision Systems

TRPO-Network Arichitecture

1. Actor Network (By actor we just mean policy)

```python
class Actor(nn.Module):
    def __init__(self, obs_dim, act_dim, hidden1, hidden2):
        super(Actor, self).__init__()
        self.fc1 = nn.Linear(obs_dim, hidden1)
        self.fc2 = nn.Linear(hidden1, hidden2)
        self.fc3 = nn.Linear(hidden2, act_dim)  # for \mu
        self.fc4 = nn.Linear(hidden2, act_dim)  # for \sigma

    def forward(self, obs):
        x = torch.tanh(self.fc1(obs))
        x = torch.tanh(self.fc2(x))
        mu = self.fc3(x)
        log_sigma = self.fc4(x)
        sigma = torch.exp(log_sigma)
        return mu, sigma

    def log_prob(self, obs, act):
        mu, sigma = self.forward(obs)
        act_distribution = Independent(Normal(mu, sigma), 1)
        log_prob = act_distribution.log_prob(act)
        return log_prob
```

Why $\log \sigma(s)$ instead of $\sigma(s)$?
$\Longrightarrow$ We have a constraint $\sigma(s) > 0$!

gives $\log \pi(a^i|s^i)$'s for (s^i, a^i) pairs

Learning and Decision Systems

TRPO-Network Arichitecture

```python
class Critic(nn.Module):
    # critic V(s ; \theta)
    def __init__(self, obs_dim, hidden1, hidden2):
        super(Critic, self).__init__()
        self.fc1 = nn.Linear(obs_dim, hidden1)
        self.fc2 = nn.Linear(hidden1, hidden2)
        self.fc3 = nn.Linear(hidden2, 1)

    def forward(self, obs):
        x = torch.tanh(self.fc1(obs))
        x = torch.tanh(self.fc2(x))

        return self.fc3(x)
```

2. Critic Network : super-easy!

Learning and Decision Systems

TRPO-Action Selection by Agent

```python
class TRPOAgent:
    def __init__(self, obs_dim, act_dim, hidden1=64, hidden2=32):
        ...

    def act(self, obs):
        obs = torch.tensor(obs, dtype=torch.float).to(device)
        with torch.no_grad():
            mu, sigma = self.pi(obs)
            act_distribution = Independent(Normal(mu, sigma), 1)
            action = act_distribution.sample()
            log_prob = act_distribution.log_prob(action)
            val = self.V(obs)
        action = action.cpu().numpy()
        log_prob = log_prob.cpu().numpy()
        val = val.cpu().numpy()

        return action, log_prob, val
```

Gaussian distribution $\mathcal{N}(\mu(s^i), \sigma(s^i))$

$a^i \sim \pi_\phi(s^i) := \mathcal{N}(\mu(s^i), \sigma(s^i))$

$\log \pi_\phi(a^i | s^i)$

$V(s^i; \theta)$

Learning and Decision Systems

TRPO-Complete Implementation

```
def train(env, agent, max_iter, gamma=0.99, lr=3e-4, lam=0.95, delta=1e-3, steps_per_epoch=4000, eval_interval=4000):
    ...
    for epoch in range(num_epochs):
        state = env.reset()
        step_count = 0
        ep_reward = 0
        for t in range(steps_per_epoch):
            action, log_prob, v = agent.act(state)
            next_state, reward, done, _ = env.step(action)
            memory.append(state, action, reward, v, log_prob)
            ep_reward += reward
            step_count += 1
            if (step_count == max_ep_len) or (t == steps_per_epoch - 1):
                s_last = torch.tensor(next_state, dtype=torch.float).to(device)
                v_last = agent.V(s_last).item()
                memory.compute_values(v_last)
            elif done:
                v_last = 0.0
                memory.compute_values(v_last)
```

LDS
Learning and Decision Systems

TRPO-Complete Implementation

```
                state = next_state
                if done:
                    state = env.reset()
                    step_count = 0
                    ep_reward = 0
                total_t += 1
            update(agent, memory, critic_optim, delta, num_updates=1)
        return
```

Learning and Decision Systems

PPO - Review

Learning and Decision Systems

Recap : TRPO

$$
\begin{aligned}
\max_{\theta} \quad & L_{\theta_{old}}(\theta) \\
\text{s.t.} \quad & \underbrace{\mathbb{E}^{\pi_{\theta_{old}}}[D_{KL}(\pi_{\theta_{old}} \parallel \pi_\theta)] \leq \delta}_{\text{approximate trust region}}
\end{aligned}
$$

$$
L_{\theta_{old}}(\theta) := \mathbb{E}_t\Big[\underbrace{\frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}}_{:= r_t(\theta)} \gamma^t A^{\pi_{\theta_{old}}}(s_t, a_t)\Big]
$$

Learning and Decision Systems

Proximal Policy Optimization : Idea

$$\begin{aligned} \max_{\theta} \quad & L_{\theta_{old}}(\theta) \\ \text{s.t.} \quad & \underbrace{\mathbb{E}^{\pi_{\theta_{old}}}[D_{KL}(\pi_{\theta_{old}} \parallel \pi_{\theta})] \leq \delta}_{\text{approximate trust region}} \end{aligned}$$

$$L_{\theta_{old}}(\theta) := \mathbb{E}_t\Big[\underbrace{\frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}}_{:= r_t(\theta)} \gamma^t A^{\pi_{\theta_{old}}}(s_t, a_t)\Big]$$

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t\Big[\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t)\Big]$$

Learning and Decision Systems

Proximal Policy Optimization : Idea

$$\boxed{\begin{array}{ll} \max\limits_{\theta} & L_{\theta_{old}}(\theta) \\ \text{s.t.} & \underbrace{\mathbb{E}^{\pi_{\theta_{old}}}[D_{KL}(\pi_{\theta_{old}} \parallel \pi_\theta)] \leq \delta}_{\text{approximate trust region}} \end{array}}$$

$$L_{\theta_{old}}(\theta) := \mathbb{E}_t\Big[\underbrace{\frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}}_{:= r_t(\theta)} \gamma^t A^{\pi_{\theta_{old}}}(s_t, a_t)\Big]$$

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t\Big[\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t)\Big]$$

removes the incentive for moving ratio outside of the interval $[1-\epsilon, 1+\epsilon]$

$\approx$ KL penalty

Learning and Decision Systems

Proximal Policy Optimization(PPO) : What's different?

```
for i in range(num_iter):
      log_probs, ent = self.pi.compute_log_prob(states, actions)
      r = torch.exp(log_probs - old_log_probs)
      clipped_r = torch.clamp(r, 1 - self.epsilon, 1 + self.epsilon)
      single_step_obj = torch.min(r * A, clipped_r * A)
      pi_loss = -torch.mean(single_step_obj)
      v = self.V(states)
      V_loss = torch.mean((v - target_v) ** 2)
      ent_bonus = torch.mean(ent)
      loss = pi_loss + 0.5 * V_loss - 0.01 * ent_bonus
      self.optimizer.zero_grad()
      loss.backward()
      self.optimizer.step()
      if i == num_iter - 1:
          kl = torch.mean(old_log_probs - log_probs).item()
          if kl > 1.5 * self.kl_threshold:
              break
    return
```

simple, surrogate loss function for policy network parameters!

$\Longrightarrow$ 1^{st}-order optimization (no Hessian needed)

Learning and Decision Systems

Thank you!

Learning and Decision Systems